Applied Data Mining

— Week 2-3—

Spring 2026

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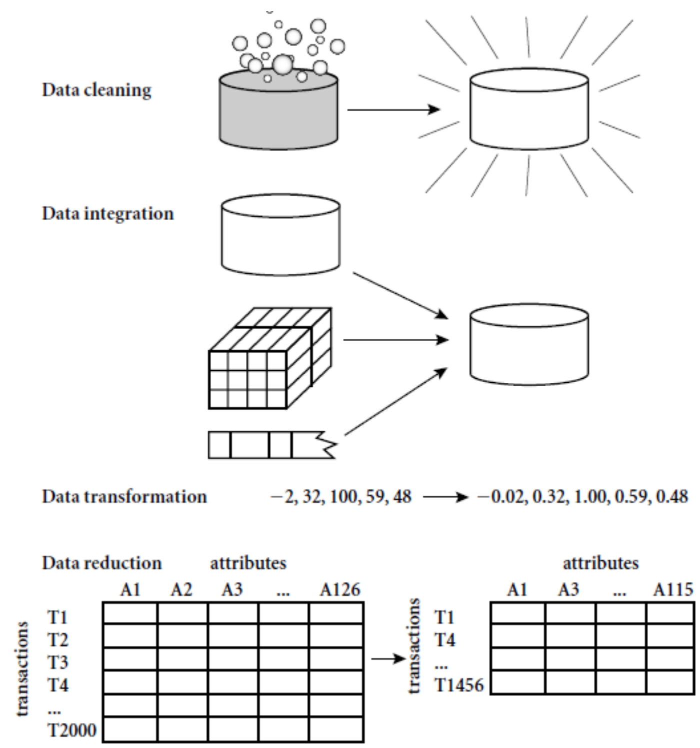
Outline

- Data Preprocessing: Definition, Goal
- Why do we preprocess the data?
- Major Tasks in Data Preprocessing
  - Data Cleaning
  - Data Integration
  - Data Reduction
  - Data Transformation



Data Preprocessing

- **What is pre-processing?**
  - Data preprocessing is the process of cleaning and transforming raw data into a suitable format for analysis, typically involving several steps, to ensure the quality and consistency of the data before using it in further analysis or modeling.

- **Goal**
  - To improve the accuracy and efficiency of downstream analysis by preparing data for machine learning algorithms or other analytical tasks.

Why Preprocess the Data?

- **Improves model performance:**
  - Cleaner and standardized data leads to better model training and prediction accuracy.
- **Reduces bias:**
  - By handling inconsistencies, preprocessing minimizes the impact of biased data on analysis results.
- **Enhances interpretability:**
  - Well-structured data makes it easier to understand and interpret analysis outcomes.

Outline

- Data Objects, Attributes, Attribute types
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Major Tasks in Data Preprocessing

- **Data cleaning**
  - Fill in missing values, smooth noisy data, identify or remove outliers, and resolve inconsistencies
- **Data integration**
  - Integration of multiple databases, data cubes, or files
- **Data reduction**
  - Dimensionality reduction
  - Data compression
- **Data transformation**
  - Normalization

Data Cleaning

- Data in the Real World Is Dirty: Lots of potentially incorrect data, e.g., instrument faulty, human or computer error, transmission error
  - incomplete: lacking attribute values, lacking certain attributes of interest, or containing only aggregate data
    - e.g., Occupation=" " (missing data)
  - noisy: containing noise, errors, or outliers
    - e.g., Salary="−10" (an error)
  - inconsistent: containing discrepancies in codes or names, e.g.,
    - Age="42", Birthday="03/07/2010"
    - Was rating "1, 2, 3", now rating "A, B, C"
    - discrepancy between duplicate records
  - Intentional (e.g., disguised missing data)
    - Jan. 1 as everyone's birthday?

Incomplete (Missing) Data

- Data is not always available
  - E.g., many tuples have no recorded value for several attributes, such as customer income in sales data
- Missing data may be due to
  - equipment malfunction
  - inconsistent with other recorded data and thus deleted
  - data not entered due to misunderstanding
  - certain data may not be considered important at the time of entry
  - not register history or changes of the data
- Missing data may need to be inferred

How to Handle Missing Data?

- **Ignore the tuple:** usually done when class label is missing (when doing classification)—not effective when the % of missing values per attribute varies considerably
- **Fill in the missing value manually:** tedious + infeasible?
- **Fill in it automatically with**
  - a global constant : e.g., "unknown", a new class?!
  - the attribute mean
  - the attribute mean for all samples belonging to the same class: smarter
  - the most probable value: inference-based such as Bayesian formula or decision tree

Noisy Data

- Noise: random error or variance in a measured variable
- Incorrect attribute values may be due to
  - faulty data collection instruments
  - data entry problems
  - data transmission problems
  - technology limitation
  - inconsistency in naming convention
- Other data problems which require data cleaning
  - duplicate records
  - incomplete data
  - inconsistent data

How to Handle Noisy Data?

- Binning
  - first sort data and partition into (equal-frequency) bins
  - then one can smooth by bin means, smooth by bin median, smooth by bin boundaries, etc.
- Regression
  - smooth by fitting the data into regression functions
- Clustering
  - detect and remove outliers

How to Handle Noisy Data?

Binning

Sorted data for *price* (in dollars): 4, 8, 15, 21, 21, 24, 25, 28, 34

Partition into (equal-frequency) bins:

Bin 1: 4, 8, 15
Bin 2: 21, 21, 24
Bin 3: 25, 28, 34

Smoothing by bin means:

Bin 1: 9, 9, 9
Bin 2: 22, 22, 22
Bin 3: 29, 29, 29

Smoothing by bin boundaries:

Bin 1: 4, 4, 15
Bin 2: 21, 21, 24
Bin 3: 25, 25, 34

4-8 , 9 , 15
= 4, 4, 4, 15

Q: Smooth the data by equal frequency bins and then apply bin boundary technique

8, 16, 9, 15, 21, 21, 24, 30, 26, 27, 34, 30.

Sort it: 8, 9, 15, 16, 21, 21, 24, 26, 27, 30, 30, 34.

frequency-4 Bins

8, 9, 15, 16

21, 21, 24, 26

27, 30, 30, 34.

Apply bin-boundary

8, 8, 16, 16

21, 21, 26, 26

~~27, 27, 30,~~

27, 27, 27, 34

frequency-3- Bins.

8, 9, 15

16, 21, 21

24, 26, 27

30, 30, 34.

Apply Bin-Boundary.

8, 8, 15

16, 21, 21

24, 27, 27

30, 30, 34.

Any Question